Ericsson PE Node (YEL-PE-11) SOP for Routing Protocol Flap (ALARM_8340)

Node: YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_8340

Description

The BGP route churn, OSPF adjacency loss, and ISIS LSP flooding failure domain on the Ericsson PE Node (YEL-PE-11) occurs due to the inherent complexities of routing protocols operating in a distributed manner across multiple nodes. Specifically, this failure domain arises from the interactions between the BGP routing protocol, which relies on neighbors to exchange routing information, and the OSPF adjacency loss or ISIS LSP flooding, which can cause instability in the routing table. This instability can be exacerbated by hardware or software issues within the PE node, such as faulty interfaces or configuration errors. In the context of the Ericsson equipment, the CLI dialect used to diagnose and resolve this failure domain must adhere to the vendor-specific syntax and commands, which are designed to provide detailed insights into the control and data planes of the network.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Yelahanka Sub-station for on-site assistance, if necessary.

- Check the status of the redundant links to ensure that they are functioning properly and can take over in the event of a failure.

- Review the network configuration and ensure that it is consistent with the vendor's recommendations and industry best practices.

Potential Root Causes

Cause	Probability
Hardware failure in the PE node	High
Software configuration error in the PE node	Medium
Routing protocol misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command displays a summary of the BGP routing table, including the number of peers and the number of routes. It helps to identify any issues with the BGP neighbor connections or the routing table itself.

Command: show ospf interface, Expected Output: Interface Ethernet1/0 is up, line protocol is up, Description: This command displays the OSPF interface status, including the interface state and the line protocol status. It helps to identify any issues with the OSPF
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adjacency or the interface itself.

Command: show isis database, Expected Output: IS-IS Level-2 Database, Description: This command displays the ISIS LSP database, including the LSP ID, sequence number, and age. It helps to identify any issues with the ISIS LSP flooding or the LSP database itself.
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Resolution Steps

Step 1: Step: Verify the PE node hardware and software configuration, Command: show version, Description: This command displays the PE node hardware and software configuration, including the version of the operating system and the firmware. It helps to identify any issues with the hardware or software configuration.

Step 2: Step: Check the PE node interfaces for any issues, Command: show interface, Description: This command displays the PE node interface status, including the interface state and the line protocol status. It helps to identify any issues with the interface itself.

Step 3: Step: Verify the routing protocol configuration, Command: show running-config | include bgp|ospf|isis, Description: This command displays the routing protocol configuration, including the BGP, OSPF, and ISIS configuration. It helps to identify any issues with the routing protocol configuration.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command displays a summary of the BGP routing table, including the number of peers and the number of routes. It helps to verify that the BGP routing table is stable and consistent.

Command: show ospf interface, Expected Output: Interface Ethernet1/0 is up, line protocol is up, Description: This command displays the OSPF interface status, including the interface state and the line protocol status. It helps to verify that the OSPF adjacency is stable and consistent.

Command: show isis database, Expected Output: IS-IS Level-2 Database, Description: This command displays the ISIS LSP database, including the LSP ID, sequence number, and age. It helps to verify that the ISIS LSP flooding is stable and consistent.
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Roll-back Procedures

- Step: Restore the original PE node configuration, Command: copy running-config startup-config, Description: This command restores the original PE node configuration from the startup-config file. It helps to roll back any changes made during the resolution process.

- Step: Verify the PE node configuration, Command: show running-config, Description: This command displays the current PE node configuration, including any changes made during the resolution process. It helps to verify that the PE node configuration is consistent with the original configuration.

- Step: Verify the routing protocol configuration, Command: show running-config | include bgp|ospf|isis, Description: This command displays the routing protocol configuration, including the BGP, OSPF, and ISIS configuration. It helps to verify that the routing protocol configuration is consistent with the original configuration.